EXP4: Build A Linear Regression Model using Python.

1. Splitting Training Data & Test Data.
2. Evaluate the model.
3. visualize the Training set & Testing set.
4. predicting test set result.
5. Compare actual output values with predicted values.

Aim:- To build a linear regression model for a dataset in ML using python. A data is stored in CSV file include splitting training & test data, Evaluate the model, visualize training & test dataset, predict test set result & compare actual output values with predicted values.

1. Splitting Training Data & Test Data:-

```
import pandas as pd
import numpy as np.
import matplotlib.pyplot as plt.
from sklearn linear_model import linear_regression
from sklearn model_selection import train_test_split

x = np.array(range(100))
y = 2.5 + 1 * (np.random.rand(len(x)).
data set = pd.DataFrame({'x':x, 'y':y})
dataset.head()
x = dataset[['x']]
y = dataset[y].
```

```
X_train, X_test, y_train, y_test = train_test_split
        (X, y, test_size = 0.2, random_state = 42).
model = LinearRegression()
model.fit(X_train, y_train).
```

2) Evaluate the model.

```
intercept = model.intercept
slope = model.coef_[0].

print ("Intercept", intercept).
print ("Slope :", slope)
```

3). Visualise Training set and Testing set.

```
plt.scatter (X_train, y_train, label = "Training Data")
plt.plot (X_train, model.predict (X_train), color =
        'Red', label = 'Regression line').
plot.ylabel ('y')
plt.xlabel ('x')
plt.legend()
plt.show().
```

Testing set

```
plt.scatter (X_test, y_test, label = "Testing Data").
plt.show (X_test, model.predict (X_test, color = 'red', label =
        Regression line").
plt.ylabel ('y')
plt.xlabel ('x')
plt.legend()
plt.show().
```

4.) predicting Test set result.

```
y_pred = model.predict(x_test)
print("predicted value : ", y_pred)
```

5) Compare Actual output value with predicted value

```
Comparision = pd.DataFrame({
        'Actual': y_test Value, 'predicted': y_pred.
        })
print(Comparision)
```